

1 QUESTION

Explain the basic construction of an n-channel enhancement MOSFET with a neat diagram. Identify the source, drain, gate, substrate, oxide layer, and channel region.

1.1 DIAGRAM

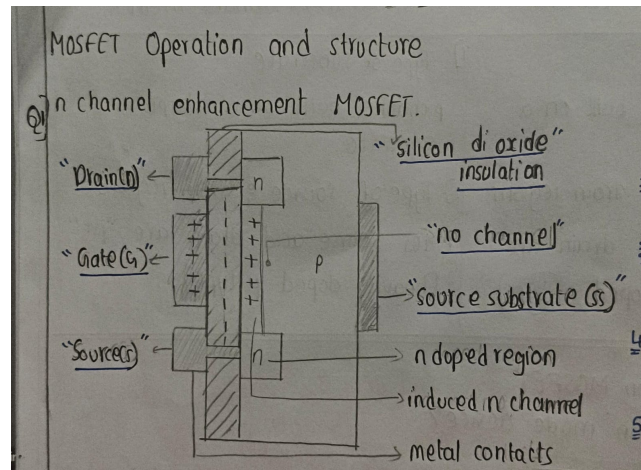


Figure 1: n channel enhancement mosfet

1.2 Explanation of the Structure of Mosfet

p-type source drain in n-type substrate, gate insulated by oxide.

- **EM:** No channel at $V_{GS} = 0$; transistor turns ON when $V_{GS} < V_T$ (negative).
- **DM:** Channel exists at $V_{GS} = 0$; transistor turns ON when $V_{GS} \leq 0$.

2 QUESTION

Explain the basic construction of a p channel enhancement mosfet. how does it differ structurally from an n channel enhancement mosfet?

2.1 DIAGRAM

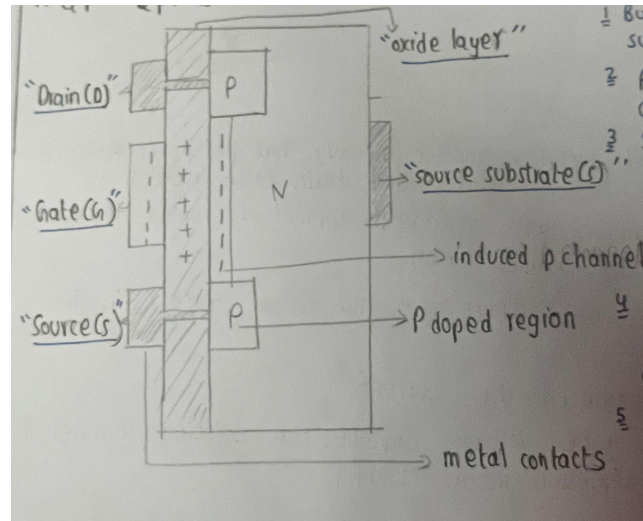


Figure 2: p channel enhancement mosfet

2.2 Explanation of the Structure of Mosfet

A p channel enhancement type mosfet often referred to as a pmos is formed by two p plus type region on the top of the n type substrate. The p regions offer low resistances. initially there is no channel exist but after the negative voltage flow p type channel is induced that allow the current to flow

2.3 how the n and p channel enhancement mosfets are structurally different?

The major difference between the pmos and nmos is their operation 1. pmos requires the negative voltage

$$V_{GS} < V_T$$

to form the channel 2. the current is carried by the holes instead of the electrons so the polarity of the voltage and the current direction is altered compared to the n channel mosfet

3 QUESTION

With the help of a diagram, explain the structure of an n-channel depletion MOSFET. Why is it called a depletion-mode device?

3.1 DIAGRAM

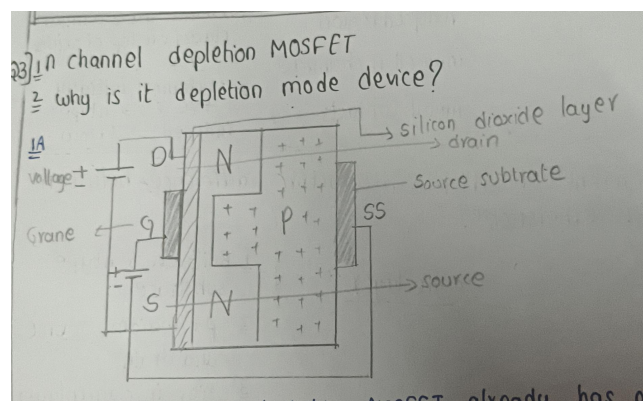


Figure 3: n channel depletion mosfet

3.2 Why is it called a depletion mode device

An n channel depletion mosfet already has pre performed n type channel between source and drain. when a negative gate voltage is applied it repels the electrons from the channel region. This reduces or depletes the no of carries in the channel there by the gate voltage depletes the existing channel it is called depletion mode mosfet.

4 QUESTION

Compare enhancement MOSFET and depletion MOSFET in terms of: (1) channel presence at $V_{GS} = 0$, (2) symbol, (3) operating mode, (4) transfer characteristics.

4.1 ANSWER

(1) channel presence at $V_{GS} = 0$

Enhancement mosfet: no channel is present.

Depletion mosfet :channel already present.

(2) symbol

Enhancement mosfet: dotted line channel

Depletion mosfet: solid line channel

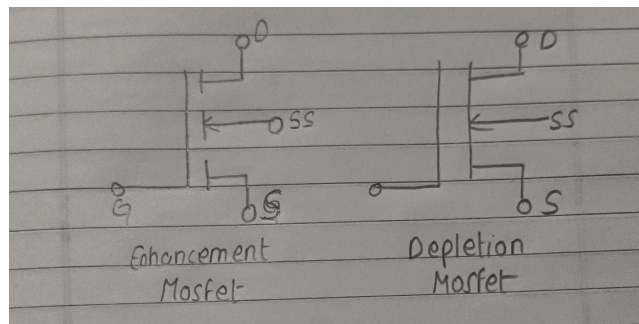


Figure 4: SYMBOLS

(3) operating mode

Enhancement mosfet: operates only in enhancement mode

Depletion mosfet: operates in both depletion and enhancement modes

(4) transfer characteristics

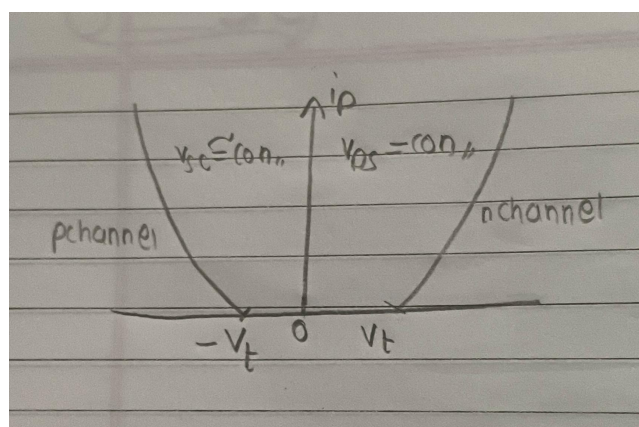


Figure 5: transfer characteristics

5 QUESTION

Explain why the gate current of an ideal MOSFET is approximately zero. What is the role of the silicon dioxide layer?

5.1 why the gate current is 0?

In an ideal mosfet the gate terminal is electrically insulated from the channel. because of this no direction conduction e path exists between the gate and the semi conductor so when a voltage is applied at the gate it creates a electric filed but no i flows into the gate. Hence the gate current is approximately 0.

5.2 role of the silicon dioxide laYer

It acts as an insulator which seperates the gate from the channel forms a capacitor and enables the voltage control also provides high input impedence.

6 QUESTION

State the condition for the formation of an inversion layer in an n-channel enhancement MOSFET. What is the significance of threshold voltage V_T ?

6.1 SCondition for inversion layer

An inversion layer is formed when the gate to source voltage exceeds the threshold value.

$$V(GS) > V_T$$

At this point, electrons accumulate at surface of ptype substrate forming a conducting n-type channel.

6.2 Significance of Threshold Voltage

It is the minimum gate voltage required to form the channel.

Below V_T MOSFET is OFF

Above V_T MOSFET is ON

It acts as a switching point in mosfet operation

7 QUESTION

Draw and explain the output characteristics of an n-channel enhancement MOSFET for different values of V_{GS} .

7.1 Diagram

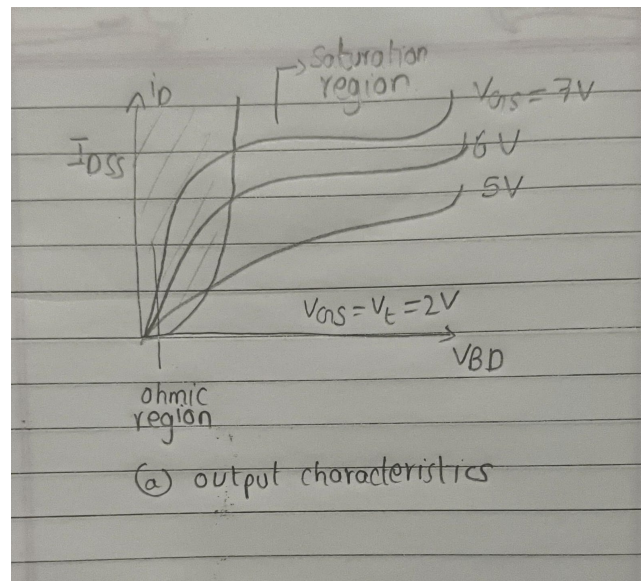
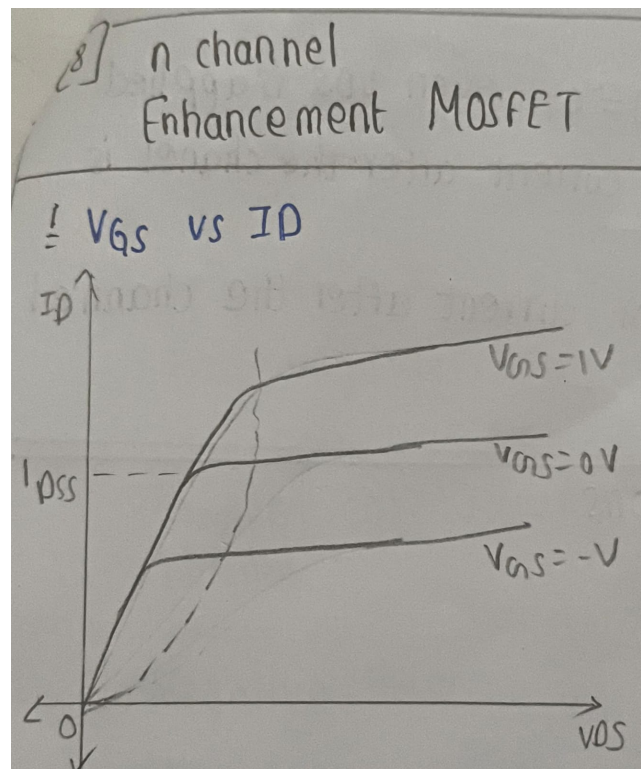


Figure 6: output characteristics

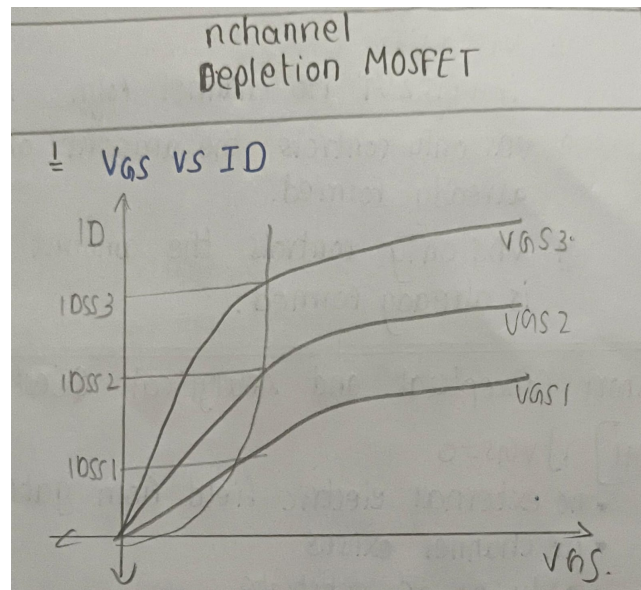
8 QUESTION

Draw and explain the transfer characteristics I_D versus V_{GS} of: (1) n-channel enhancement MOSFET, (2) n-channel depletion MOSFET.

8.1 n channel enhancement mosfet



n channel depletion mosfet



9 QUESTION

Distinguish clearly between the ohmic (triode) region, saturation region, and cutoff region of an n-channel MOSFET.

9.1 DIFFERENCES

Region	Condition	channel	ID	Behavior
Cutoff Region	$V_{GS} < V_T$	No channel	$I_D \approx 0$	MOSFET is OFF
ohmic Region	$V_{GS} > V_T$ $V_{DS} < V_{GS} - V_T$	channel formed	$I_D \uparrow$ $V_{DS} \uparrow$	Acts like a variable resistor
Saturation	$V_{DS} = V_{GS} - V_T$	channel pinched	$I_D \approx \text{constant}$	Acts like a current source.

10 QUESTION

student says: "Since MOSFET is voltage controlled, V_{DS} alone determines whether the device conducts." Is this statement correct? Justify.

10.1 ANSWER

MOSFET is called a voltage controlled device because it controls whether the device conducts or not. If $V_{GS} > V_T$, no channel forms, $I_D = 0$ even V_{DS} is applied. V_{DS} only controls the amount of current after the channel is already formed.

11 QUESTION

For an n-channel enhancement MOSFET, explain what happens physically when: (1)

$$V_{GS} = 0$$

(2)

$$0 < V_{GS} < V_T$$

(3)

$$V_{GS} > V_T$$

11.1 $V_{GS} = 0$

No channel no current flow

11.2 $0 < V_{GS} < V_T$

weak inversion layer is formed so small current will flow

11.3 $V_{GS} > V_T$

strong channel will form that results large current flow

12 QUESTION

Why is a depletion MOSFET able to conduct even when $V_{GS} = 0$, whereas an enhancement MOSFET cannot?

12.1 ANSWER

A depletion MOSFET already has a channel, so current can flow even when $V_{GS}=0$.

13 QUESTION

For an n-channel depletion MOSFET, explain the effect of: (1) negative V_{GS} , (2) positive V_{GS} , on drain current.

13.1 ANSWER

For an n channel depletion mosfet Negative V_{GS} reduces the channel width, so drain current decreases. Positive V_{GS} increases the channel width, so drain current increases.

14 QUESTION

For a p-channel MOSFET, explain the polarity requirements of V_{GS} and V_{DS} for normal operation.

ANSWER

For a p-channel MOSFET, V_{GS} must be negative to turn it ON. V_{ds} must also be negative for current flow.
This states normal operation with current flowing from source to drain.

15 QUESTION

Write the drain current equation of an enhancement MOSFET in saturation: $I_D = K(V_{GS} - V_T)^2$ and explain the meaning of each term. Also state the condition under which this equation is valid

ANSWER

The drain current equation in saturation for an enhancement MOSFET is:

$$I_D = K(V_{GS} - V_T)^2$$

where I_D is the drain current, V_{GS} is the gate-to-source voltage, V_T is the threshold voltage, and K is a constant related to the device.

This equation is valid when $V_{GS} > V_T$ and $V_{DS} \geq (V_{GS} - V_T)$. A source resistor creates a voltage based on the drain current. This makes VGS adjust automatically when current changes. If current increases, VGS becomes more negative and brings it back down

16 QUESTION

Design a fixed-bias circuit for an n-channel enhancement MOSFET with: $V_{DD} = 12\text{ V}$, $V_T = 2\text{ V}$, $K = 0.5\text{ mA/V}^2$ such that the drain current is approximately 2 mA. Find the required VGS.

ANSWER

The drain current equation in saturation is:

$$I_D = K(V_{GS} - V_T)^2$$

Given:

$$I_D = 2\text{ mA}, \quad K = 0.5\text{ mA/V}^2, \quad V_T = 2\text{ V}$$

Substituting:

$$2 = 0.5(V_{GS} - 2)^2$$

$$(V_{GS} - 2)^2 = 4$$

$$V_{GS} - 2 = 2$$

$$V_{GS} = 4\text{ V}$$

17 QUESTION

A source resistor bias circuit is used for an n-channel depletion MOSFET. Explain how self-bias can stabilize the operating point.

ANSWER

A source resistor creates a voltage based on the drain current. This makes VGS adjust automatically when current changes. If current increases, VGS becomes more negative and brings it back down

18 QUESTION

Design a simple voltage-divider gate bias network for an n-channel enhancement MOS- FET so that the gate voltage is 4 V from a 12 V supply. Choose practical resistor values.

ANSWER

A simple voltage-divider gate bias network can be designed using:

$$V_G = V_{DD} \cdot \frac{R_2}{R_1 + R_2}$$

Given: $V_G = 4\text{ V}$ and $V_{DD} = 12\text{ V}$, we get the ratio:

$$\frac{R_2}{R_1 + R_2} = \frac{4}{12} = \frac{1}{3} \Rightarrow R_2 : R_1 = 1 : 2$$

Choosing resistor values

$$R_1 = 200\text{ k}\Omega, \quad R_2 = 100\text{ k}\Omega$$

19 QUESTION

Explain why voltage-divider bias is generally better than fixed gate bias for maintaining a stable Q-point in MOSFET circuits.

ANSWER

Voltage-divider bias keeps the gate voltage steady on its own. It handles changes in the MOSFET better. Fixed bias can let the Q-point drift. So voltage-divider bias keeps the circuit stable.

20 QUESTION

An n-channel enhancement MOSFET amplifier is to be biased at the midpoint of the drain characteristics for maximum symmetrical swing. Explain the design philosophy for choosing R_D and the Q-point.

ANSWER

The Q-point is chosen at the midpoint of the drain characteristics for maximum symmetrical swing.

Drain voltage at Q-point:

$$V_D = \frac{V_{DD}}{2}$$

The drain resistor is calculated using Ohm's law:

$$R_D = \frac{V_{DD} - V_D}{I_D}$$

This ensures the MOSFET stays in saturation:

$$V_{DS} \geq V_{GS} - V_T$$

Thus, the amplifier reaches the maximum undistorted output swing.

21 QUESTION

An n-channel enhancement MOSFET has: $V_T = 2\text{ V}$, $K = 0.8\text{ mA/V}^2$ If $V_{GS} = 5\text{ V}$, find the drain current I_D in saturation.

SOLUTION

1. $I_D = K(V_{GS} - V_T)^2$
2. $I_D = 0.8(5 - 2)^2$
3. $I_D = 0.8(3)^2$
4. $I_D = 0.8 \times 9$
5. $I_D = 7.2\text{ mA}$

22 QUESTION

For the MOSFET in Q21, if $V_{DS} = 4\text{ V}$, determine whether the device operates in saturation or ohmic region.

SOLUTION

1. Given: $V_{GS} = 5\text{ V}$, $V_T = 2\text{ V}$, $V_{DS} = 4\text{ V}$
2. $V_{GS} - V_T = 5 - 2 = 3\text{ V}$
3. Condition: $V_{DS} \geq (V_{GS} - V_T)$
4. Check: $4 \geq 3$
5. Therefore, MOSFET operates in **saturation region**

23 QUESTION

23 An n-channel enhancement MOSFET has: $V_T = 1.5 \text{ V}$, $K = 1 \text{ mA/V}^2$, $V_{DD} = 15 \text{ V}$, $R_D = 2 \text{ k}\Omega$ and the gate is fixed at $V_G = 4.5 \text{ V}$, $V_S = 0$. Find: (1) V_{GS} , (2) I_D , (3) V_D , (4) V_{DS} .

SOLUTION

1. Given: $V_T = 1.5 \text{ V}$, $K = 1 \text{ mA/V}^2$, $V_{DD} = 15 \text{ V}$, $R_D = 2 \text{ k}\Omega$
2. $V_G = 4.5 \text{ V}$, $V_S = 0$
3. $V_{GS} = V_G - V_S = 4.5 - 0 = 4.5 \text{ V}$
4. Assume saturation: $I_D = K(V_{GS} - V_T)^2$
5. $I_D = 1(4.5 - 1.5)^2 = 1(3)^2 = 9 \text{ mA}$
6. $V_D = V_{DD} - I_D R_D = 15 - (9 \times 2) = -3 \text{ V}$
7. Since V_D is negative, assumption is wrong
8. Device operates in **ohmic (triode) region**
9. Solving gives: $V_{DS} = 3 \text{ V}$, $I_D = 6 \text{ mA}$
10. Final: $V_{GS} = 4.5 \text{ V}$, $V_D = 3 \text{ V}$, $V_{DS} = 3 \text{ V}$

24 QUESTION

24 An n-channel depletion MOSFET has $I_{DSS} = 10 \text{ mA}$, $V_{GS(off)} = 4 \text{ V}$. Using the transfer relation find I_D for: (1) $V_{GS} = 0 \text{ V}$, (2) $V_{GS} = 2 \text{ V}$, (3) $V_{GS} = +1 \text{ V}$.

SOLUTION

1. Given: $I_{DSS} = 10 \text{ mA}$, $V_{GS(off)} = -4 \text{ V}$
2. Formula: $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_{GS(off)}}\right)^2$
3. Case 1: $V_{GS} = 0 \text{ V}$
4. $I_D = 10 \left(1 - \frac{0}{-4}\right)^2 = 10(1)^2 = 10 \text{ mA}$
5. Case 2: $V_{GS} = -2 \text{ V}$
6. $I_D = 10 \left(1 - \frac{-2}{-4}\right)^2 = 10(1 - 0.5)^2 = 10(0.5)^2 = 2.5 \text{ mA}$
7. Case 3: $V_{GS} = 1 \text{ V}$
8. $I_D = 10 \left(1 - \frac{1}{-4}\right)^2 = 10(1 + 0.25)^2 = 10(1.25)^2$
9. $I_D = 10 \times 1.5625 = 15.625 \text{ mA}$
10. Final: $I_D = 10 \text{ mA}$, 2.5 mA , 15.625 mA

25 QUESTION

An n-channel enhancement MOSFET is biased by a source resistor. Given: $V_{DD} = 18 \text{ V}$, $R_D = 3 \text{ k}\Omega$, $R_S = 1 \text{ k}\Omega$, $V_G = 5 \text{ V}$, $V_T = 2 \text{ V}$, $K = 1 \text{ mA/V}^2$ find the operating point (I_D , V_{DS}). Bias circuit for Q25

SOLUTION

1. Given: $V_{DD} = 18\text{ V}$, $R_D = 3\text{ k}\Omega$, $R_S = 1\text{ k}\Omega$, $V_G = 5\text{ V}$, $V_T = 2\text{ V}$, $K = 1\text{ mA/V}^2$

2. $V_{GS} = V_G - V_S = 5 - I_D R_S = 5 - I_D$

3. Using $I_D = K(V_{GS} - V_T)^2$:

$$I_D = (5 - I_D - 2)^2 = (3 - I_D)^2$$

4.

$$I_D = 9 - 6I_D + I_D^2$$

5.

$$I_D^2 - 7I_D + 9 = 0$$

6.

$$I_D = \frac{7 \pm \sqrt{49 - 36}}{2} = \frac{7 \pm \sqrt{13}}{2}$$

7. $I_D = \frac{7 \pm 3.6}{2} \Rightarrow I_D = 5.3\text{ mA or } 1.7\text{ mA}$

8. $I_D = 1.7\text{ mA}$

9. $V_D = 18 - (1.7 \times 3) = 12.9\text{ V}$, $V_S = 1.7\text{ V}$

10. $V_{DS} = V_D - V_S = 12.9 - 1.7 = 11.2\text{ V}$

26 QUESTION

An enhancement MOSFET has $V_T = 2\text{ V}$, $K = 0.5\text{ mA/V}^2$. At what value of V_{GS} will the drain current be 4.5 mA ?

SOLUTION

1. Given: $V_T = 2\text{ V}$, $K = 0.5\text{ mA/V}^2$, $I_D = 4.5\text{ mA}$

2. Using $I_D = K(V_{GS} - V_T)^2$

3. $4.5 = 0.5(V_{GS} - 2)^2$

4. $9 = (V_{GS} - 2)^2$

5. $V_{GS} - 2 = \pm 3$

6. $V_{GS} = 5\text{ V or } -1\text{ V}$

7. Taking valid value: $V_{GS} = 5\text{ V}$

27 QUESTION

A MOSFET has $V_{DD} = 20\text{ V}$, $R_D = 4\text{ k}\Omega$, $I_D = 2\text{ mA}$. Find the drain voltage V_D . If the source is grounded, find V_{DS} .

SOLUTION

1. Given: $V_{DD} = 20\text{ V}$, $R_D = 4\text{ k}\Omega$, $I_D = 2\text{ mA}$

2. $V_D = V_{DD} - I_D R_D$

3. $V_D = 20 - (2 \times 4) = 20 - 8 = 12\text{ V}$

4. Since source is grounded, $V_S = 0$

5. $V_{DS} = V_D - V_S = 12 - 0 = 12\text{ V}$

28 QUESTION

An n-channel enhancement MOSFET is biased with $V_G = 6\text{ V}$, $R_S = 1.5\text{ k}\Omega$, $V_T = 2\text{ V}$, $K = 0.5\text{ mA/V}^2$. Find I_D and V_S .

SOLUTION

1. Given: $V_G = 6\text{ V}$, $R_S = 1.5\text{ k}\Omega$, $V_T = 2\text{ V}$, $K = 0.5\text{ mA/V}^2$

2. $V_{GS} = V_G - V_S = 6 - I_D R_S = 6 - 1.5 I_D$

3. Using $I_D = K(V_{GS} - V_T)^2$:

$$I_D = 0.5(6 - 1.5 I_D - 2)^2 = 0.5(4 - 1.5 I_D)^2$$

4.

$$I_D = 0.5(16 - 12 I_D + 2.25 I_D^2) = 8 - 6 I_D + 1.125 I_D^2$$

5.

$$1.125 I_D^2 - 7 I_D + 8 = 0$$

6.

$$I_D = \frac{7 \pm \sqrt{49 - 36}}{2.25} = \frac{7 \pm \sqrt{13}}{2.25}$$

7. $I_D = 4.71\text{ mA}$ or 1.51 mA

8. Checking validity: $V_{GS} > V_T$ gives valid $I_D = 1.51\text{ mA}$

9. $V_S = I_D R_S = 1.51 \times 1.5 \approx 2.27\text{ V}$

10. Final: $I_D \approx 1.51\text{ mA}$, $V_S \approx 2.27\text{ V}$

29 QUESTION

In a voltage-divider bias circuit, the gate is connected to two resistors: $R_1 = 1\text{ M}\Omega$, $R_2 = 500\text{ k}\Omega$, $V_{DD} = 15\text{ V}$. Find the gate voltage V_G .

SOLUTION

1. Given: $R_1 = 1\text{ M}\Omega$, $R_2 = 500\text{ k}\Omega$, $V_{DD} = 15\text{ V}$

2. Using voltage divider:

$$V_G = V_{DD} \cdot \frac{R_2}{R_1 + R_2}$$

3. $V_G = 15 \cdot \frac{500}{1000 + 500}$

4. $V_G = 15 \cdot \frac{500}{1500} = 15 \cdot \frac{1}{3}$

5. $V_G = 5\text{ V}$

30 QUESTION

An n-channel enhancement MOSFET is required to operate in saturation with: $I_D = 3\text{ mA}$, $V_{DD} = 12\text{ V}$, $V_D = 6\text{ V}$. Find the value of drain resistor R_D .

SOLUTION

1. Given: $I_D = 3\text{ mA}$, $V_{DD} = 12\text{ V}$, $V_D = 6\text{ V}$

2. Using: $V_D = V_{DD} - I_D R_D$

3. $6 = 12 - (3\text{ mA}) \cdot R_D$

4. $(3\text{ mA}) \cdot R_D = 6$

5. $R_D = \frac{6}{3\text{ mA}} = 2\text{ k}\Omega$